

ECO 403 Macroeconomics

Final term-Current Paper #2023

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ECO403

Current Paper #2023

IS Curve Slope

The IS curve represents the equilibrium in the goods market. Its slope is negative, showing the inverse relationship between interest rates and real output. As interest rates decrease, investment increases, leading to higher output.

LM Curve Slope

The LM curve represents the equilibrium in the money market. Its slope is positive, indicating the relationship between interest rates and real income. Higher income leads to higher money demand, requiring higher interest rates to maintain equilibrium.

Budget Constraint

The budget constraint shows the combinations of goods an individual or entity can afford given their income and prices. It helps determine feasible consumption choices.

LM Curve Graph

The LM curve is upward sloping, indicating the relationship between interest rates and real income where the money market is in equilibrium.

Indifference Curve

An indifference curve shows various combinations of two goods that provide the same level of satisfaction for an individual. It helps analyze consumer preferences and choices.

Currency M1 and M2

M1 refers to the most liquid components of the money supply, like physical currency and demand deposits.

M2 includes M1 along with less liquid assets like savings accounts and small time deposits.

Solow Growth Golden Rule

The Solow growth model determines the optimal saving rate for steady state growth. The golden rule saving rate maximizes consumption for future generations.

Population Growth and Technological Progress Equations

In the Solow model, population growth and technological progress are often represented as exogenous variables affecting economic growth. The equations depict their impact on output per capita.

IS Curve, Exchange Rate, Output Level

Changes in the exchange rate can impact net exports, causing shifts in the IS curve and affecting the equilibrium output level.

Tobin's Formula

$Q = \text{Market Value of Installed Capital} / \text{Replacement Cost of Installed Capital}$

Tobin's formula relates the demand for money to the opportunity cost of holding money (which is the interest rate) and the risk of holding money.

C < Y and C > Y Explanation

C < Y refers to consumption being less than output, indicating saving.

C > Y means consumption exceeds output, leading to dis saving, which could be financed by drawing on past savings or borrowing.

Short Run Consumption vs. Long Run Consumption

In the short run, consumption can be influenced by current income and temporary factors. In the long run, consumption is guided by permanent income and economic factors.

Supply Curve Diagram and Explanation

A supply curve in macroeconomics typically represents the relationship between the price level and the level of aggregate output supplied in an economy. It usually has a positive slope due to factors like production costs and profit considerations.

Policy, Risk Premium, and Investment

If a policy increases the risk premium (extra return for holding risky assets), it can lead to higher interest rates, which might discourage investment due to higher borrowing costs.

Commodity Money vs. Fiat Money

Commodity Money is physical money that has intrinsic value, such as gold or silver.

Fiat Money is money declared by a government to have value, but it lacks intrinsic value and is backed by trust.

Business Fixed Investment vs. Residential Fixed Investment

Business Fixed Investment refers to spending by businesses on capital goods like machinery or buildings.

Residential Fixed Investment refers to spending on residential structures like homes and apartment buildings.

Three Sectors Government, Firms, Households

Government Collects taxes, provides public goods, and influences economic activity through policies.

Firms Produce goods and services, generate income, and create employment.

Households Consume goods and services, provide labor, and save money.

Inflation vs. Hyperinflation

Inflation is a general increase in prices over time.

Hyperinflation is an extreme form of inflation where prices skyrocket uncontrollably, often due to a collapse in a country's currency.

Efficiency vs. Equity

Efficiency aims to maximize overall economic output, often resulting in unequal distribution of resources.

Equity focuses on fairness and reducing disparities in income and wealth, potentially impacting overall efficiency.

Interest Rate vs. Investment

Interest Rate is the cost of borrowing or the return on saving.

Investment is spending on capital goods that enhance production, affected by interest rates; higher Rates may discourage investment.

Relative Price Distortion

Relative Price Distortion refers to a situation where changes in prices disrupt the usual patterns of supply and demand in an economy, leading to misallocation of resources.

Proving Net Export ($NX = EX - IM$)

Net Exports (NX) equals Exports (EX) minus Imports (IM). This equation represents the net value of a country's trade balance.

Inflation Growth Rate Relationship

Inflation and growth are macroeconomic concepts. Inflation studies are more common in macroeconomics, while growth studies are prevalent in both macroeconomics and microeconomics.

Budget Deficits in Solow Growth Model

Budget deficits can impact the Solow model by affecting public saving and thus the investment rate. Higher deficits can reduce national saving and lead to lower investment and growth in the long run.

Permanent Income & Transitory Income

Permanent income is the consistent level of income that individuals expect over time. **Transitory income** is temporary or irregular income fluctuations. Consumption decisions are more influenced by permanent income.

LM Shocks

LM shocks refer to changes in the money market equilibrium due to shifts in the money supply or money demand. They affect interest rates and can influence overall economic activity.

IS LM Model

The IS LM model analyzes the interaction between the goods market (IS curve) and the money market (LM curve) to determine equilibrium interest rates and output levels.

IS LM Supply Topic

This topic likely discussed how changes in the supply side of the economy (e.g., technology, labor force) can impact the IS and LM equilibrium and influence macroeconomic outcomes.

MPK Formula Components

Marginal Product of Capital (MPK) formula components include changes in output (ΔY) and changes in capital (ΔK), representing how additional capital contributes to output growth.

Solow Growth Model

The Solow model explains economic growth by considering factors like capital accumulation, population growth, and technological progress. It provides insights into long term economic development.

Aggregate Demand Downward Sloping

The downward sloping aggregate demand curve results from the wealth effect, interest rate effect, and foreign trade effect. As price levels rise, real wealth decreases, interest rates

increase, and net exports decrease, leading to lower overall demand.

Income Effect vs. Substitution Effect

Income Effect Change in quantity demanded due to a change in real income, holding prices constant.

Substitution Effect Change in quantity demanded due to a change in relative prices, assuming constant real income.

Adaptive Expectations vs. Rational Expectations

Adaptive Expectations Forming future expectations based on past experiences, with adjustments over time.

Rational Expectations Forming expectations by considering all available information, assuming agents are rational and make unbiased forecasts.

Corporate Income Tax vs. Investment Tax Credit

Corporate Income Tax Tax on a corporation's profits, which can reduce post tax profits and potentially affect investment decisions.

Investment Tax Credit A tax incentive that reduces taxes based on the amount invested in certain assets, encouraging investment.

IS LM and AD AS Curve Explanation

IS LM Curve Shows the equilibrium in the goods and money markets, indicating combinations of interest rates and output where both markets are in balance.

AD AS Curve (Aggregate Demand Aggregate Supply) Illustrates the equilibrium between overall demand and supply in an economy, helping analyze output and price levels.

Golden Rule of Solow Model

The Golden Rule suggests that a country should save and invest at a rate that maximizes consumption per worker in the long run, ensuring sustainable and optimal economic growth.

Solow Growth Model Equation

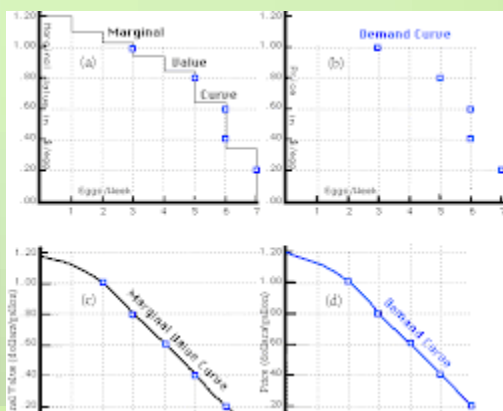
The Solow model's basic equation is $Y = A \cdot F(K, L)$, where Y is output, A is total factor productivity, K is capital, and L is labor. It shows how output depends on capital, labor, and productivity.

Difference Between Commodity and Fiat Money

Commodity Money Physical money with intrinsic value (e.g., gold, silver) beyond its use as a medium of exchange.

Fiat Money Currency declared by a government as legal tender, without intrinsic value, relying on trust and government backing.

Graph between Marginal Value and Quantity



Graph of marginal value and quantity

The graph of marginal value and quantity is a downward sloping curve that indicates that as the quantity of a good or service increases, the marginal value of each additional unit decreases. This is due to the law of diminishing marginal utility, where people tend to get less satisfaction from each additional unit, and the increased opportunity cost of each additional unit as more of the good or service is produced. This graph can be used to illustrate economic concepts like consumer surplus and producer surplus, which are the difference between the

amount consumers are willing to pay for a good or service and the actual amount they pay. It can also be used to analyze market equilibrium, where the marginal value of a good or service equals the marginal cost of producing it.

GDP Deflator

Long Explanation The GDP deflator is a measure of the overall price level in an economy, representing the ratio of nominal GDP to real GDP, multiplied by 100.

Short Explanation GDP deflator measures price changes in an economy, reflecting inflation or deflation.

Rising Unemployment Rate in Pakistan

The unemployment rate in Pakistan can rise due to factors such as rapid population growth, limited job opportunities, economic instability, and insufficient skill development.

Natural Unemployment Rate

The natural unemployment rate is the rate at which the economy operates at full employment without causing excessive inflation. Its value depends on structural factors and can be found using economic data.

M1 and M2

M1 and M2 are classifications of the money supply

M1 includes physical currency, demand deposits, and traveler's checks.

M2 includes M1 plus savings accounts, time deposits, and some other near money assets.

Types of Debt (Floating, Unfunded, Permanent)

Floating Debt Short term debt that the government can quickly change (e.g., treasury bills).

Unfunded Debt Obligations without dedicated funds (e.g., pensions, Social Security).

Permanent Debt Debt with no maturity date, typically in the form of bonds.

Steady State Level of Capital per Worker

The steady state level of capital per worker occurs when investment equals depreciation, leading to a constant capital to labor ratio.

Production Smoothing Motive for Holding Inventories

The production smoothing motive aims to maintain a steady level of production even if demand fluctuates. Holding inventories helps bridge the gap between production and demand, ensuring smoother operations.

Liquidity Preference Theory and Demand Curve for Real Money Balances

The liquidity preference theory suggests people demand money for transactions and precautionary reasons. The demand curve for real money balances shows the relationship between the interest rate and the quantity of money people wish to hold.

Phillips Curve Equation

The Phillips Curve represents a tradeoff between inflation and unemployment. The equation generally shows an inverse relationship between the two variables.

Impact on Average Propensity to Consume

Changes in income can influence the Average Propensity to Consume (APC). As income rises, APC usually decreases because people tend to save a larger proportion of their income.

Aggregate Demand Curve

The Aggregate Demand Curve shows the total demand for goods and services at different price levels in an economy. It slopes downward due to the wealth effect, interest rate effect, and foreign trade effect.

Policies to Increase Saving Rate

- Tax Incentives for Saving
- Public Awareness Campaigns
- Social Security Reforms
- Retirement Plans Promotion
- Education on Financial Literacy

Mundell Fleming Model Equilibrium with Graph

The Mundell Fleming model analyzes the interaction between exchange rates, interest rates, and output. Equilibrium occurs where the IS curve intersects the LM curve in the open economy.

Impact on Real Rental Price

Factors such as changes in demand, supply, inflation, and market conditions can influence real rental prices.

Use the golden rule of capital in the Solow growth model and determine whether the saving rate and capital stock are too high or too low(3)

The golden rule of capital in the Solow growth model states that an economy should choose a savings rate that maximizes consumption per capita in the long run. To determine whether the saving rate and capital stock are too high or too low:

If the current savings rate is higher than the golden rule savings rate the economy is saving too much, leading to lower consumption and a higher capital stock.

If the current savings rate is lower than the golden rule savings rate the economy is saving too little, resulting in higher consumption and a lower capital stock.

Discuss any three reasons how the technological process can be encouraged in an economy in the context of Solow growth model (3)

Encouraging technological progress in the context of the Solow growth model can be done through:

Research and development (R&D) incentives: Providing subsidies or tax incentives for firms to invest in R&D

Education and human capital development: Investing in education and skill development to improve the workforce's ability to innovate.

Intellectual property protection: Strengthening laws that protect intellectual property rights, encouraging firms to invest in innovation.

Difference between fixed exchange rate and floating exchange rate. (3)

Fixed exchange rate: A fixed exchange rate is when a country's currency is tied to a specific value, often another currency like the US dollar. The exchange rate remains constant and is maintained through central bank interventions.

Floating exchange rate: A floating exchange rate is determined by the foreign exchange market based on supply and demand. The value of the currency fluctuates freely, and the central bank may not intervene to maintain a specific value.

Calculate the amount budget deficit was over stated by using the given information
Budgeted deficit 27000, inflation rate 7% Government debt 175000 (3) check this

To calculate the overstated budget deficit due to inflation:

Overstated deficit = Actual deficit $\times$ (1 + Inflation rate) Actual deficit

Overstated deficit = $27000 \times (1 + 0.07)$ $27000 = 1890$

Difference the average propensity to consume and marginal propensity to consume (3)

Average Propensity to Consume (APC) is the ratio of total consumption to total income.
Marginal Propensity to Consume (MPC) is the change in consumption resulting from a change in income. The main difference is that APC is a cumulative measure, while MPC is an incremental measure.

Formula of money multiplier(3)

Making substitutions for the fractions above, we obtain: $M = \frac{cr + 1}{cr + rr} \times B$

The monetary base B is the total number of dollars held by the public as currency C and by the banks as reserves R. • The reservedeposit ratio rr is the fraction of deposits D that banks hold in reserve R. • The currencydeposit ratio cr is the amount of currency C people hold as a fraction of their holdings of demand deposits D.

An Example Suppose, monetary base B is \$500 billion, the reserve deposit ratio rr is 0.1 and currency deposit ratio cr is 0.6 The money multiplier is: $m = \frac{0.6 + 1}{0.6 + 0.1} = 2.3$ And the money supply is: $M = 2.3 \times \$ 500 \text{ billion} = \$1,150 \text{ billion}$

Three reasons why the firm hold inventories? (5)

Reasons why firms hold inventories:

Production Smoothing: To maintain steady production levels even when demand fluctuates.

Transaction Motive: To have enough inventory to meet daily operational needs.

Speculative Motive: In anticipation of future price changes or shortages.

Five policies increase saving rate in economy(5)

Policies to increase saving rate:

Tax incentives for savings

Retirement saving plans with employer contributions

Financial literacy programs
Government saving schemes
Encouraging longterm investments

Mundell Fleming model is given discuss IS curve in import and export if government impose tariff and quota on exports (5)

In the MundellFleming model, if the government imposes tariffs and quotas on exports:

The IS curve shifts left due to reduced exports, leading to lower income and output.

The impact on imports and the balance of payments depends on the size of the tariffs and quotas.

How decrease in tax rate can affect the aggregate expenditure and nominal level in the economy (5)

A decrease in tax rates can affect aggregate expenditure and the nominal level in the economy:

Aggregate Expenditure: Lower tax rates increase disposable income, leading to higher consumption and aggregate expenditure.

Nominal Level: Increased consumption can lead to higher demand, potentially increasing prices and nominal GDP.

1. Equilibrating Aggregate Demand Curve (5 marks):

In the aggregate demand (AD) curve, equilibrium occurs where AD equals the aggregate supply (AS) or output. On a graph, this is where the AD curve intersects the AS curve. This intersection point represents the level of output where planned spending (AD) matches the economy's total production (AS), ensuring no unplanned inventory changes. Equilibrium output is the point where the two curves intersect.

2. Concept of Residential Investment (3 marks):

Residential investment refers to the expenditures made by households and businesses on residential structures, such as houses and apartments. It includes the construction, renovation, and improvement of these structures. Residential investment contributes to economic growth by creating jobs in the construction sector and stimulating demand for related goods and services.

3. Components of National Identity per Worker in the Solow Model (3 marks):

In the Solow model, the national income identity per worker is expressed as: $Y/L = C/L + I/L + G/L$. Here:

Y/L : Output per worker.

C/L: Consumption per worker.

I/L: Investment per worker.

G/L: Government spending per worker.

4. Consumer Savings and C less than Y, C greater than Y (3 marks):

If C (consumption) is less than Y (income/output), it indicates that households are saving more than they are spending, which leads to an increase in savings. This can affect economic growth in the short term due to reduced immediate consumption.

If C is greater than Y, it implies that households are spending more than their income, which can be sustained through borrowing or drawing down on savings. This could lead to shortterm economic growth due to increased consumption but might raise concerns about longterm sustainability.

5. Calculating Marginal Product of Capital (MPK) with Graph (5 marks):

The marginal product of capital (MPK) represents the additional output produced when one more unit of capital is added while holding other factors constant. On a graph, MPK is illustrated by the slope of the production function curve. As capital increases, the slope (MPK) typically diminishes, reflecting diminishing returns to capital.

6. Impact on Aggregate Demand and Output in ISLM Model (3 marks):

In the ISLM model:

An increase in aggregate demand (AD) will lead to a higher output level (Y) due to the interaction of the goods market (IS) and the money market (LM).

A decrease in AD will result in lower output. The ISLM model demonstrates how changes in interest rates and income levels influence aggregate demand and, consequently, output.

Differentiate between DemandPull Inflation and CostPush Inflation (5 marks):

1. DemandPull Inflation:

Occurs when aggregate demand (total spending) in an economy exceeds its aggregate supply (total production) at full employment. This leads to increased demand for goods and services, driving up their prices. It's typically a result of strong consumer and business spending, often fueled by increased money supply or government policies.

2. CostPush Inflation:

Arises when production costs for goods and services increase, leading to higher prices. This can result from factors like rising wages, increased raw material costs, or supply disruptions. Costpush inflation can reduce real output and might be accompanied by decreased consumer purchasing power.

Short Questions:

1. Ways to Encourage Technological Progress:

Research and Development (R&D) Funding: Invest in R&D to develop new technologies and improve existing ones.

Education and Training: Enhance workforce skills and knowledge to adapt to technological changes.

Intellectual Property Protection: Ensure strong patent and copyright laws to incentivize innovation.

2. Effect of Increased Government Spending (G) on IS Curve and Output Level:

Increased government spending shifts the IS (InvestmentSavings) curve to the right, raising equilibrium output level. This is due to higher total spending in the economy resulting from increased government expenditures.

3. Effects of Increasing Saving Rate:

Increasing the saving rate leads to lower consumption and higher savings. In the short term, it can decrease aggregate demand and economic growth. However, in the long term, higher savings can finance investment, leading to increased production and potential economic growth.

4. Properties of Consumption Function in Keynesian Theory:

Marginal Propensity to Consume (MPC): The change in consumption resulting from a change in income. It's positive but less than 1.

Autonomous Consumption: The consumption level when income is zero, reflecting essential spending.

Average Propensity to Consume (APC): The ratio of consumption to income, indicating the portion of income spent.

5. Impact of Marginal Product of Capital on Investment Function:

As the marginal product of capital (MPK) decreases, the incentive for investment diminishes. A higher MPK implies that investing in more capital would yield greater output, thus encouraging investment. Conversely, a lower MPK might lead to reduced investment enthusiasm.

Additional Short Questions:

1. Consumption Function and Its Terms:

The consumption function represents the relationship between disposable income and consumption spending by households.

$$C = C_0 + cY_d$$

C: Consumption, C_0 : Autonomous consumption, c: Marginal propensity to consume, Y_d : Disposable income.

2. Equation of Motion for Solow Model:

The Solow model's equation of motion describes how capital stock (K) changes over time:

$$\Delta K / \Delta t = sY - (\delta + n + g)K$$

s: Saving rate, Y: Output, δ : Depreciation rate, n: Population growth rate, g: Technological progress rate.

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Let's ace those studies together!

Feel free to ask me about any queries

Remember me in your prayers

☆ ★ ☆ ★ ☆ Wish you all the best ☆ ★ ☆ ★ ☆

